Document Identifier: **PMC8989534**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: The skin sample was obtained from blepharoplasty of the indicated healthy donor under the ethical approval of the Peking Union Medical College Hospital Institutional Review
Field Sample Permit: The mice used in this study were purchased from SPF Biotechnology Co . , Ltd. (Beijing) and reared at 25 ± 1°C under 12-h light–dark cycle with food and water fed ad libitum . hMPC implantation assay hMPC implantation assay was conducted as previously described (48,49,61,62) .
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
Human primary fibroblasts were derived from the upper eyelid skin of a 28-year-old male individual and cultured on 0.1 % gelatin-precoated dishes with fibroblast culture medium (high glucose DMEM [Hyclone] supplemented with 10 % fetal bovine serum , 1 % penicillin/streptomycin and 0.1 mM NEAAs) following previous protocols (55) .
Subject Demographics
Age: Human primary fibroblasts were derived from the upper eyelid skin of a 28-year-old male individual and cultured on 0.1 % gelatin-precoated dishes with fibroblast culture medium (high glucose DMEM [Hyclone] supplemented with 10 % fetal bovine serum , 1 % penicillin/streptomycin and 0.1 mM NEAAs) following previous protocols (55) .
Randomization
The percentage of SA-β-gal-positive cells was counted in randomly selected fields of view for each replicate using ImageJ software (NIH) .
Blinding
not detected.
Power Analysis

not detected.	
Replication	
Replicates for each sample were merged for downstream analysis .	
Cell Line Authentication	
Authentication: hESC clones were then genotyped by genomic PCR and sequencing .	
Code Information	
Identifiers: https://github.com/dinovski/asDamID/blob/master/scripts/hmmt_functions.R).	https://github.com/dinovski/asDamID/blob/master/scripts/hmmt_functions.R
Data Information	
Identifiers: Primary cynomolgus monkey MPCs (cynMPCs) were also seeded on 0.1 % gelatin-precoated dishes and cultured in MPC culture medium supplemented with 100 nM tenofovir (BCP02416 , BiochemPartner)	BCP02416
Identifiers: The mice used in this study were purchased from SPF Biotechnology Co . , Ltd. (Beijing) and reared at 25 ± 1°C under 12-h light–dark cycle with food and water fed ad libitum . hMPC implantation assay hMPC implantation assay was conducted as previously described (48,49,61,62) .	48,49,61,62

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Subsequently , cells were washed with PBS , and incubated with secondary antibodies and Hoechst 33342 (H3570 , Thermo Fisher Scientific , 1:1,000) in 10 % donkey serum for 1 h at room temperature , and then washed with PBS before being mounted with antifade mounting medium (H-1000 , Vector Labs) .	antifade mounting medium (H-1000 , Vector Labs) .		
For Co-IP with exogenous proteins in HEK293T cells , the supernatants were incubated with FLAG antibody-coupled beads (A2220 , Sigma) and rotated at 4°C overnight .	FLAG		
After sonication , samples were centrifugated at 12,000 × g for 10 min at 4°C and the resulting supernatants were then incubated with Protein A Dynabeads (Thermo Fisher Scientific) pre-conjugated with anti-BMAL1 , anti-H3K9me3 or anti-IgG antibodies .	anti-BMAL1		
	anti-H3K9me3		
	anti-IgG		
Experimental Models: Cell Lines			
MATERIALS AND METHODS Cell culture BMAL1+/+ human embryonic stem cells (hESCs, line H9, WiCell Research Institute) and BMAL1–/– hESCs were seeded on feeder cells, a layer of mitomycin C-inactivated mouse embryonic fibroblasts (MEFs), and cultured in hESC culture medium containing DMEM/F12 (Thermo Fisher Scientific) supplemented with 20	H9		
HEK293T cells transfected with plasmids expressing FLAG-Luc , FLAG-BMAL1 (WT) and FLAG-BMAL1 (Δ88) , as well as hMPCs and cynMPCs , were lysed using CHAPS lysis buffer (120 mM NaCl , 1 mM EDTA , 40 mM HEPES [pH 7.5] , 0.3 % CHAPS and 1 × complete protease inhibitor cocktail [Roche]) at 4°C for 5 h and were centrifuged at 14,500 × g at 4°C for 45 min .	HEK293T		Suggestion: RRID:CVCL_0063(link)
Experimental Models: Organisms/Strains			

In Brief , approximately 3 × 106 BMAL1+/+ or BMAL1 –/– hESCs were resuspended in Matrigel: mTeSR (1:4) solution and injected into the inguinal region of male NOD-SCID mice (6–8 weeks old) .	NOD-SCID		
Recombinant DNA			
Briefly , a guide RNA targeting exon 8 of BMAL1 was cloned into a pCAGmCherry-gRNA vector (#87110 , Addgene) , marking the plasmid as BMAL1-gRNA-mCherry .	pCAGmCherry-gRNA		Suggestion: RRID:Addgene_87110(link)
Next , BMAL1+/+ hESCs were resuspended in the plasmid cocktail that contained BMAL1-gRNA-mCherry and pCAG-1BPNLS-Cas9-1BPNLS-2AGFP plasmid (#87109 , Addgene) in Opti-MEM (Thermo Fisher Scientific) .	pCAG-1BPNLS-Cas9-1BPNLS-2AGFP		Suggestion: RRID:Addgene_87109(link)
After PCR products purification by the PCR Purification Kit (Qiagen) , the OCT4 promoter fragments were cloned into the pMD20 T vector (TAKARA) and sequenced.	pMD20 T		
In brief , cDNA fragments amplified using primers listed in Supplementary Table S1 were cloned into the pLE4 vector (a gift from Tomoaki Hishida) .	pLE4		
BMAL1-targeting sgRNA (sg-BMAL1) or non-targeting control (NTC) sgRNA (sg-NTC) was cloned into the lentiCRISPRv2 plasmid (#52961 , Addgene) , which carries a hSpCas9 expression cassette and a U6-driven sgRNA cassette. shRNA targeting STING or GL2 was cloned into the pLVTHM plasmid (#12247 , Addgene)	lentiCRISPRv2		
	pLVTHM		
HEK293T cells were transfected with the plasmids and lentiviral vectors (psPAX2 (#12260 , Addgene) and pMD2G (#12259 , Addgene)) using Lipofectamine 3000 (Invitrogen) .	psPAX2		
	pMD2G		
To prepare sgRNA and SpCas9 mRNAs for embryo injection , T7-cynBMAL1 sgRNA and T7-SpCas9-expressing plasmids were linearized and in vitro transcribed with MEGAscript T7 and mMESSAGE mMACHINE T7 ULTRA Kits (Thermo Fisher Scientific) .	T7-SpCas9-expressing		
Software and Algorithms			

The adipocytes , osteoblasts and chondrocytes that differentiated from hMPCs were correspondingly characterized by Oil Red O staining (O1391 , Sigma) and FABP4 immunostaining (indicating adipogenesis) , von Kossa staining (GMS 80045.3 , GenMed Scientific Inc . , U.S.A .) , and Osteocalcin staining (indicating osteogenesis) and toluidine blue (T3260 , Sigma) (indicating chondrogenesis) following the manufacturer's instructions .	Sigma		
Immunoreactive bands were then detected by a ChemiDoc XRS + system (Bio-Rad) and statistically quantified by ImageJ software (NIH) .	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
The 3D reconstruction of H3K9me3 staining was performed by serial z-stack sectioning for up to 30 images at 100-nm intervals using a Leica SP5 confocal system , and further processed to perform 3D reconstruction by Imaris software (version 7.4.2) as previously described (71) .	Imaris		
We searched data against UniProt human protein database (updated on 02–2016) .	UniProt		Suggestion: UniProt (RRID:SCR_002380)(link)
FDR (false discovery rate) analysis by Percolator was set to < 1 % for protein identification , and the peptide confidence was set as high for peptide filter .	Percolator		
For ChIP-seq , library construction was conducted using KAPA Hyper Prep Kit (KAPA Biosystems , KK8504) and Index Primers Set 1 for Illumina (New England Biolabs (NEB) , E7335L) following the manufacturer's instructions .	ChIP-seq		
RNA-seq data processing Raw paired-end reads were trimmed by Trim Galore software (version 0.4.5) (https://github.com/FelixKrueger/TrimGalore).	https://github.com/FelixKrueger/TrimGalore		
Then , the cleaned reads were mapped to the cynomolgus macaque (Macaca Fascicularis) MacFas5.0 genome obtained from UCSC genome browser database using hisat2 (version 2.0.4) (86) .	UCSC genome browser		
	hisat2		
Read counts for each gene were calculated by HTSeq (version	HTSeq		

0.11.0) with only high-quality mapped reads (mapping quality > 20) (87) .			
Differentially expressed genes (DEGs) between tissues derived from BMAL1+/+ and BMAL1–/– cynomolgus monkeys were calculated by R package DESeq2 (version 1.22.2) with the cutoff set to ‘the adjusted P-value < 0.05 and the absolute log2 (fold change) > 0.5’ .	DESeq2		Suggestion: DESeq, RRID:SCR_000154(link)
Gene Ontology (GO) and pathway enrichment analysis were conducted by Metascape (88) .	Metascape		
For calculating differentially expressed repetitive elements (REs) in cynMPCs and cynomolgus monkey tissues , trimmed reads were mapped to the cynomolgus monkey (Macaca Fascicularis) MacFas5.0 genome obtained from UCSC genome browser database using STAR software (version 2.5.2b) (89) .	STAR		
Then , the read counts for each RepeatMasker-annotated repeats of MacFas5.0 genome were calculated by featureCounts (version 2.0.0) (90) with only high-quality mapped reads (mapping quality > 20)	featureCounts		
DamID-seq data processing Raw reads of Dam and Dam-EMD data for BMAL1+/+ and BMAL1–/– hMPCs were trimmed by Trim Galore software (version 0.4.5) .	Trim Galore		
The DamID signals were visualized for each 2,000-base pair (bp) bin using the bamCompare program in deepTools (version 2.5.4) software (94) .	deepTools		
Duplicated reads were discarded by MarkDuplicates . jar program in Picard tools (version 1.119) .	Picard		
Then , reads were sorted with SAMtools (version 1.6) (93) .	SAMtools		Suggestion: SAMTOOLS, RRID:SCR_002105(link)
For H3K9me3 peak calling , SICER	SICER		
Then LAD regions were partitioned into ‘H3K9me3-enriched LADs’ and ‘Other LADs’ by robust k-means analysis with two clusters using R package pheatmap (version 1.0.12) .	pheatmap		Suggestion: pheatmap (RRID:SCR_016418)(link)
Genomic distribution of ATAC peaks were annotated by annotatePeaks.pl program in homer2 (version 4.10.3) software (95).	homer2		

Statistical analysis Data are presented as the means ± SEM. P-value (P) was analyzed by two-tailed Student's t-test using GraphPad Prism 6 software.	GraphPad Prism		Suggestion: GraphPad Prism (RRID:SCR_002798)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Both permutation-based P-values (P) and Benjamini–Hochberg adjusted Q values (Q) < 0.05 were considered as rhythmic .	Benjamini–Hochberg
Statistical significances were assessed by a two-tailed unpaired Student’s t-test .	t-test
Oligonucleotides	
Primers used for PCR: me-OCT4 forward: ATTTGTTTTTTGGGTAGTTAAAGGT , me-OCT4 reverse: CCAACTATCTTCATCTTAATAACATCC (59) .	CCAACTATCTTCATCTTAATAACA TCC

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